

Pesticide exposure and serum organochlorine residuals among testicular cancer patients and healthy controls.

The incidence of testicular cancer (TC) has been increasing worldwide during the last decades. The reasons of the increase remains unknown, but recent findings suggest that organochlorine pesticides (OPs) could influence the development of TC. A hospital-based case-control study of 50 cases and 48 controls was conducted to determine whether environmental exposure to OPs is associated with the risk of TC, and by measuring serum concentrations of OPs, including p,p'-dichlorodiphenyldichloroethylene (p,p'-DDE) isomer and hexachlorobenzene (HCB) in participants. A significant association was observed between TC and household insecticide use (odds ratio [OR] = 3.01, 95 % CI: 1.11-8.14; OR(adjusted) = 3.23, 95 % CI: 1.15-9.11). Crude and adjusted ORs for TC were also significantly associated with higher serum concentrations of total OPs (OR = 3.15, 95 % CI: 1.00-9.91; OR(adjusted) = 3.34, 95 % CI: 1.09-10.17) in cases compared with controls. These findings give additional support to the results of previous research that suggest that some environmental exposures to OPs may be implicated in the pathogenesis of TC.